Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: E&D Azevedo Dairy #4

Physical address of dairy:

Number and Street	City	County	Zip Code
2222 S Healy RD	Merced	Merced	95341

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X066-X110-X027-XXXX

B. OPERATORS

Azevedo, Debbie

Operator name: Azevedo, Debbie
Telephone no.: (209) 812-4171 Landline, (209) 404-3973 Cellular

Mailing Address Number and Street	City	State	Zip Code
6178 Spring Valley Drive	Atwater	CA	95301

Azevedo, Eugenio

Operator name: Azevedo, Eugenio
Telephone no.: Landline, (209) 401-3636 Cellular

Mailing Address Number and Street	City	State	Zip Code
6178 Spring Valley Drive	Atwater	CA	95301

C. OWNERS

Machado, Tony

Legal owner name: Machado, Tony
Telephone no.: Landline, (209) 761-9696 Cellular

Mailing Address Number and Street	City	State	Zip Code
2222 S. Healy Road	Merced	CA	95341

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	303	794	768	0
Number under roof	0	155	0	0	0	0
Maximum number	0	665	580	1,100	900	0
Average number	0	155	303	794	768	0
Avg live weight (lbs)	0	1,450	1,000	750		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 15,743.15 *tons per reporting period*

Total nitrogen from manure: 171,637.60 *lbs per reporting period*

After ammonia losses (30% loss applied): 120,146.32 *lbs per reporting period*

Total phosphorus from manure: 24,126.94 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 35,642.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 26,440,000 *gallons*

Total nitrogen generated: 33,202.28 *lbs*

Total phosphorus generated: 12,246.53 *lbs*

Total potassium generated: 147,079.22 *lbs*

Total salt generated: 538,532.89 *lbs*

	26,440,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	26,440,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dom-1	Ground water
Dom-2	Ground water
Dom-3	Ground water
IW-1	Ground water
IW-2	Ground water

Source Description	Type
IW-3	Ground water
MID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	86	86	2	both	X066-X110-X027-XXXX
Field 2	29	29	2	both	X066-X110-X027-XXXX
Field 3	26	26	2	both	X066-X110-X041-XXXX X066-X110-X042-XXXX
Field 4	51	51	2	both	X066-X110-X041-XXXX X066-X110-X042-XXXX
Field 5	52	52	2	both	X066-X110-X041-XXXX X066-X110-X042-XXXX
Totals for areas that were used for application	244	244	10		
Totals for areas that were not used for application					
Land application area totals	244	244	10		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/07/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 86 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	1,308.00 *ton*	Dry-weight		67.2	19,200.00	3,040.00	2,970.00		14.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	972.00
Total actual harvest content	15.21	191.56	30.33	29.63	1,416.78

Field 1

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 86 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	2,102.00 *ton*	Dry-weight		62.8	11,300.00	1,800.00	16,800.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	24.44	205.49	32.73	305.50	1,727.55

Field 2

Field name: Field 2

11/07/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 29 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	427.00 *ton*	Dry-weight		58.8	19,300.00	2,710.00	26,400.00		14.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	972.00
Total actual harvest content	14.72	234.16	32.88	320.30	1,759.24

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	620.00 *ton*	Dry-weight		63.4	11,000.00	1,670.00	15,300.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	21.38	172.15	26.13	239.44	1,471.07

Field 3

Field name: Field 3

11/08/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 26 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	344.00 *ton*	Dry-weight		67.8	20,300.00	2,900.00	28,100.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	972.00
Total actual harvest content	13.23	172.97	24.71	239.43	1,116.20

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	598.00 *ton*	Dry-weight		69.8	12,600.00	1,830.00	15,100.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	23.00	175.04	25.42	209.77	1,444.77

Field 4

Field name: Field 4

11/08/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 51 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	630.00 *ton*	Dry-weight		56.2	19,800.00	2,980.00	30,000.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	972.00
Total actual harvest content	12.35	214.26	32.25	324.64	1,417.57

Field 4

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 51 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/24/2023	1,285.00 *ton*	Dry-weight		70.6	11,700.00	2,740.00	18,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	25.20	173.34	40.59	277.05	918.55

Field 5

Field name: Field 5

11/08/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 52 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	669.00 *ton*	Dry-weight		64.2	21,000.00	2,950.00	27,700.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	972.00
Total actual harvest content	12.87	193.44	27.17	255.16	1,179.09

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/24/2023	1,308.00 *ton*	Dry-weight		68.6	12,100.00	2,710.00	18,400.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	25.15	191.14	42.81	290.66	884.61

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/07/2022: Wheat, silage, soft dough

Field name: Field 1

Crop: Wheat, silage, soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	186.80	56.67	167.85	3,059.14	688.00 *ton*
Application event totals		186.80	56.67	167.85	3,059.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	25.32	9.67	111.34	516.44	1,728,000.00 *gal*
IW-1	Ground water	0.45	0.00	0.00	156.91	9,240,000.00 *gal*
Application event totals		25.77	9.67	111.34	673.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	25.32	9.67	111.34	516.44	1,728,000.00 *gal*
Application event totals		25.32	9.67	111.34	516.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	16.85	6.70	43.76	249.00	1,296,000.00 *gal*
Application event totals		16.85	6.70	43.76	249.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-1	Ground water	0.49	0.00	0.00	171.17	10,080,000.00 *gal*
Application event totals		0.49	0.00	0.00	171.17	

Field 1 - 05/30/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	52.65	18.38	268.95	724.36	3,456,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	22.62	9,324,000.00 *gal*
Application event totals		52.65	18.38	268.95	746.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.16	9,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-1	Ground water	0.24	0.00	0.00	83.55	4,920,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	11.94	4,920,000.00 *gal*
Application event totals		0.24	0.00	0.00	95.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.16	9,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Uan-32 Fertilizer Side Dress	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Field 1 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.37	10,044,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.16	9,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-1	Ground water	0.24	0.00	0.00	83.55	4,920,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	11.94	4,920,000.00 *gal*
Application event totals		0.24	0.00	0.00	95.48	

Field 2 - 11/07/2022: Wheat, silage, soft dough

Field name: Field 2

Crop: Wheat, silage, soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	186.80	56.67	167.85	3,059.14	232.00 *ton*
Application event totals		186.80	56.67	167.85	3,059.14	

Field 2 - 11/07/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	25.03	9.56	110.06	510.51	576,000.00 *gal*
IW-2	Ground water	0.53	0.00	0.00	157.21	2,652,000.00 *gal*
Application event totals		25.55	9.56	110.06	667.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	25.03	9.56	110.06	510.51	576,000.00 *gal*
Application event totals		25.03	9.56	110.06	510.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	16.66	6.63	43.26	246.14	432,000.00 *gal*
IW-2	Ground water	0.57	0.00	0.00	169.30	2,856,000.00 *gal*
Application event totals		17.22	6.63	43.26	415.44	

Field 2 - 06/01/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	52.05	18.17	265.86	716.04	1,152,000.00 *gal*
IW-2	Ground water	0.52	0.00	0.00	154.18	2,601,000.00 *gal*
Application event totals		52.56	18.17	265.86	870.22	

Field 2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	25.68	3,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-2	Ground water	0.78	0.00	0.00	231.54	3,906,000.00 *gal*
Application event totals		0.78	0.00	0.00	231.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-2	Ground water	0.37	0.00	0.00	111.86	1,887,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	13.58	1,887,000.00 *gal*
Application event totals		0.37	0.00	0.00	125.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Uan-32 Fertilizer Side Dress	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-2	Ground water	0.37	0.00	0.00	111.86	1,887,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	13.58	1,887,000.00 *gal*
Application event totals		0.37	0.00	0.00	125.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	27.15	3,774,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.15	

Field 2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	27.88	3,876,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.88	

Field 3 - 11/08/2022: Wheat, silage, soft dough

Field name: Field 3

Crop: Wheat, silage, soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	163.45	49.59	146.87	2,676.74	182.00 *ton*
Application event totals		163.45	49.59	146.87	2,676.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	27.92	10.67	122.76	569.41	576,000.00 *gal*
IW-3	Ground water	0.40	0.00	0.00	244.26	2,520,000.00 *gal*
Application event totals		28.31	10.67	122.76	813.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	24.77	9.85	64.34	366.05	576,000.00 *gal*
Application event totals		24.77	9.85	64.34	366.05	

Field 3 - 11/08/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	21.77	7.60	111.20	299.50	432,000.00 *gal*
IW-2	Ground water	0.58	0.00	0.00	174.55	2,640,000.00 *gal*
Application event totals		22.35	7.60	111.20	474.05	

Field 3 - 06/01/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	58.05	20.26	296.54	798.66	1,152,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	22.82	2,844,000.00 *gal*
Application event totals		58.05	20.26	296.54	821.48	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	26.00	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-2	Ground water	0.37	0.00	0.00	111.08	1,680,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	13.48	1,680,000.00 *gal*
Application event totals		0.37	0.00	0.00	124.56	

Field 3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.07	3,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Uan-32 Fertilizer Side Dress	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.75	3,084,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	23.11	2,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-3	Ground water	0.23	0.00	0.00	139.58	1,440,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	11.55	1,440,000.00 *gal*
Application event totals		0.23	0.00	0.00	151.13	

Field 4 - 11/08/2022: Wheat, silage, soft dough

Field name: Field 4

Crop: Wheat, silage, soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 11/08/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	163.45	49.59	146.87	2,676.74	357.00 *ton*
Application event totals		163.45	49.59	146.87	2,676.74	

11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	32.02	12.24	140.81	653.15	1,296,000.00 *gal*
Application event totals		32.02	12.24	140.81	653.15	

11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	32.02	12.24	140.81	653.15	1,296,000.00 *gal*
IW-3	Ground water	0.36	0.00	0.00	219.40	4,440,000.00 *gal*
Application event totals		32.38	12.24	140.81	872.55	

02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	18.94	7.54	49.20	279.92	864,000.00 *gal*
Application event totals		18.94	7.54	49.20	279.92	

04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.05	5,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.05	

Field 4 - 06/04/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	66.59	23.24	340.15	916.10	2,592,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	19.14	4,680,000.00 *gal*
Application event totals		66.59	23.24	340.15	935.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-3	Ground water	0.48	0.00	0.00	298.27	6,036,000.00 *gal*
Application event totals		0.48	0.00	0.00	298.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.54	6,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Uan-32 Fertilizer Side Dress	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	25.04	6,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	25.04	6,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.04	

Field 4 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-2	Ground water	0.36	0.00	0.00	108.60	3,222,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	13.18	3,222,000.00 *gal*
Application event totals		0.36	0.00	0.00	121.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.54	6,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.54	

Field 5 - 11/08/2022: Wheat, silage, soft dough

Field name: Field 5

Crop: Wheat, silage, soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	163.45	49.59	146.87	2,676.74	364.00 *ton*
Application event totals		163.45	49.59	146.87	2,676.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	31.41	12.00	138.10	640.59	1,296,000.00 *gal*
IW-3	Ground water	0.36	0.00	0.00	221.00	4,560,000.00 *gal*
Application event totals		31.76	12.00	138.10	861.59	

Field 5 - 11/08/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	42.15	16.77	109.46	622.79	1,960,000.00 *gal*
Application event totals		42.15	16.77	109.46	622.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	21.77	7.60	111.20	299.50	864,000.00 *gal*
IW-2	Ground water	0.49	0.00	0.00	146.78	4,440,000.00 *gal*
Application event totals		22.26	7.60	111.20	446.28	

Field 5 - 06/06/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	65.31	22.79	333.60	898.49	2,592,000.00 *gal*
MID Canal	Surface water	0.00	0.00	0.00	17.33	4,320,000.00 *gal*
Application event totals		65.31	22.79	333.60	915.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	23.59	5,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.59	

Field 5 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.22	6,036,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.89	6,204,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Uan-32 Fertilizer Side Dress	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	24.55	6,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	25.37	6,324,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.37	

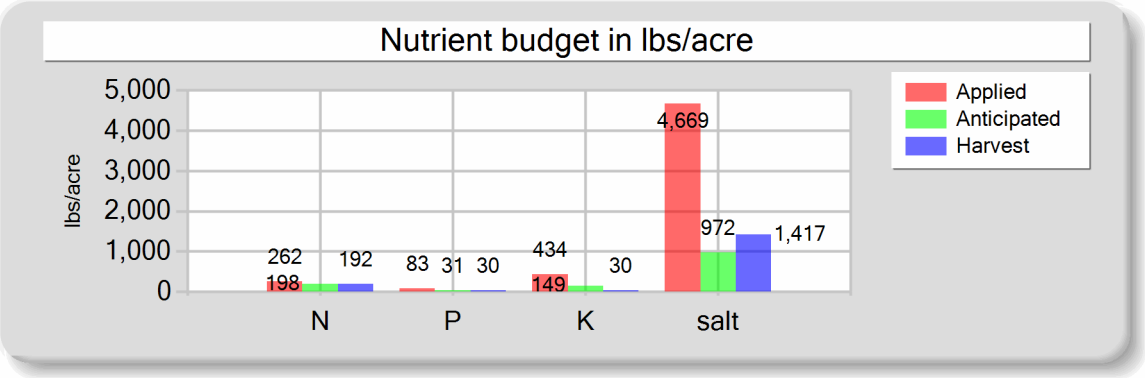
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.00	0.00	0.00	23.59	5,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.59	

B. NUTRIENT BUDGET

Field 1 - 11/07/2022: Wheat, silage, soft dough

Field name: Field 1 Crop: Wheat, silage, soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	186.80	56.67	167.85	3,059.14
Process wastewater	67.49	26.05	266.44	1,281.89
Fresh water	0.94	0.00	0.00	328.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.22	82.72	434.28	4,669.10
Anticipated crop nutrient removal	198.00	30.60	149.40	972.00
Actual crop nutrient removal	191.56	30.33	29.63	1,416.78
Nutrient balance	70.66	52.39	404.65	3,252.32
Applied to removed ratio	1.37	2.73	14.66	3.30

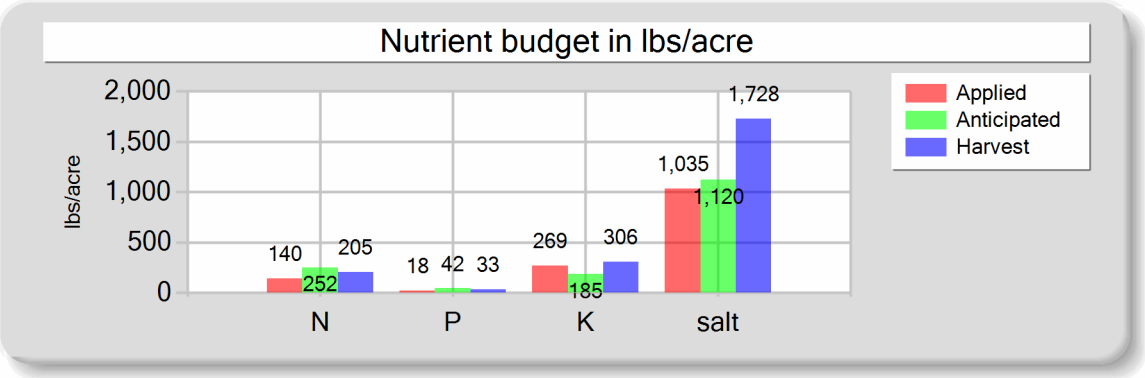
Fresh water applied	
19,320,000.00	*gallons*
711.49	*acre-inches*
8.27	*inches/acre*

Process wastewater applied	
4,752,000.00	*gallons*
175.00	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/30/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	52.65	18.38	268.95	724.36
Fresh water	0.48	0.00	0.00	310.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	140.13	18.38	268.95	1,034.80
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	205.49	32.73	305.50	1,727.55
Nutrient balance	-65.36	-14.36	-36.55	-692.76
Applied to removed ratio	0.68	0.56	0.88	0.60

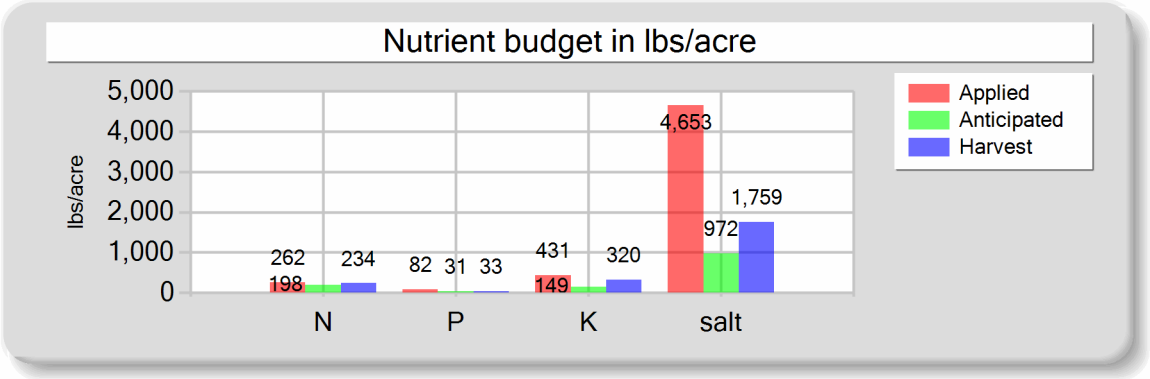
Fresh water applied	
68,928,000.00	*gallons*
2,538.38	*acre-inches*
29.52	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/07/2022: Wheat, silage, soft dough

Field name: Field 2 Crop: Wheat, silage, soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	186.80	56.67	167.85	3,059.14
Process wastewater	66.71	25.75	263.38	1,267.15
Fresh water	1.09	0.00	0.00	326.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.60	82.42	431.22	4,652.79
Anticipated crop nutrient removal	198.00	30.60	149.40	972.00
Actual crop nutrient removal	234.16	32.88	320.30	1,759.24
Nutrient balance	27.44	49.54	110.92	2,893.55
Applied to removed ratio	1.12	2.51	1.35	2.64

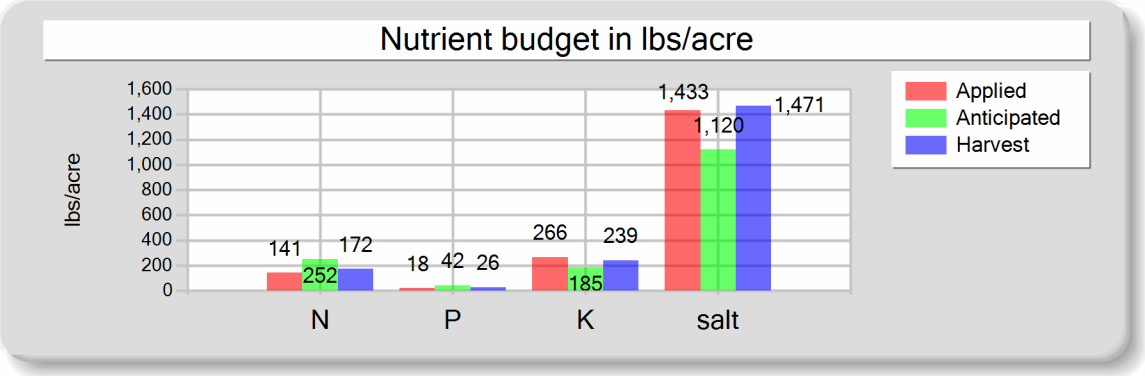
Fresh water applied	
5,508,000.00	*gallons*
202.84	*acre-inches*
6.99	*inches/acre*

Process wastewater applied	
1,584,000.00	*gallons*
58.33	*acre-inches*
2.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/01/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	52.05	18.17	265.86	716.04
Fresh water	2.04	0.00	0.00	717.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	141.09	18.17	265.86	1,433.34
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	172.15	26.13	239.44	1,471.07
Nutrient balance	-31.06	-7.97	26.42	-37.73
Applied to removed ratio	0.82	0.70	1.11	0.97

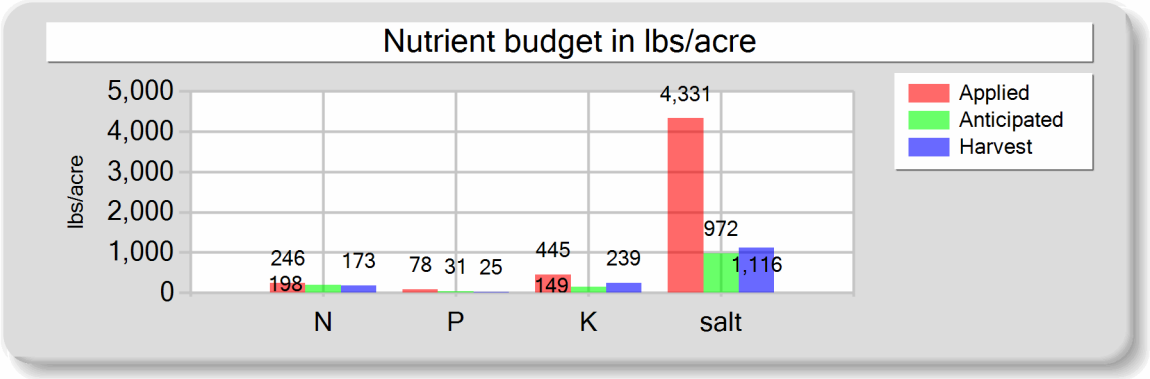
Fresh water applied	
25,275,000.00	*gallons*
930.79	*acre-inches*
32.10	*inches/acre*

Process wastewater applied	
1,152,000.00	*gallons*
42.42	*acre-inches*
1.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/08/2022: Wheat, silage, soft dough

Field name: Field 3 Crop: Wheat, silage, soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.45	49.59	146.87	2,676.74
Process wastewater	74.46	28.12	298.29	1,234.96
Fresh water	0.98	0.00	0.00	418.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.89	77.71	445.16	4,330.52
Anticipated crop nutrient removal	198.00	30.60	149.40	972.00
Actual crop nutrient removal	172.97	24.71	239.43	1,116.20
Nutrient balance	72.92	53.00	205.73	3,214.32
Applied to removed ratio	1.42	3.14	1.86	3.88

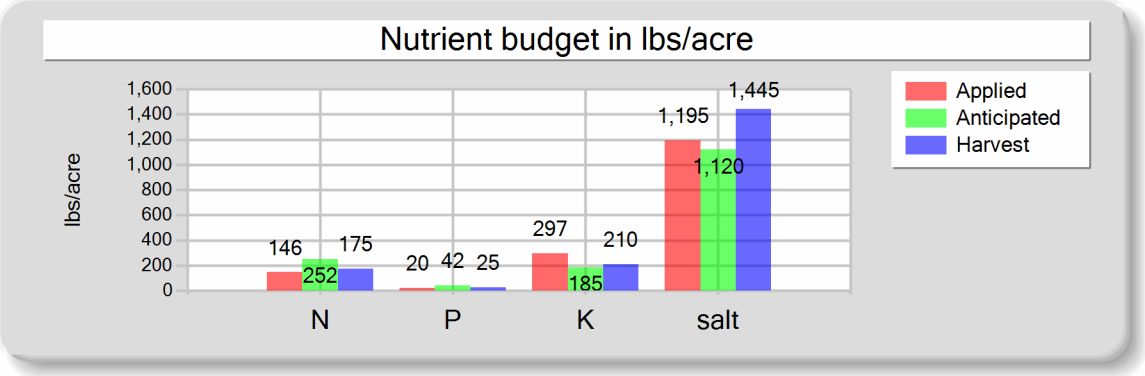
Fresh water applied	
5,160,000.00	*gallons*
190.03	*acre-inches*
7.31	*inches/acre*

Process wastewater applied	
1,584,000.00	*gallons*
58.33	*acre-inches*
2.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/01/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	58.05	20.26	296.54	798.66
Fresh water	0.60	0.00	0.00	396.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	145.65	20.26	296.54	1,195.09
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	175.04	25.42	209.77	1,444.77
Nutrient balance	-29.39	-5.16	86.77	-249.67
Applied to removed ratio	0.83	0.80	1.41	0.83

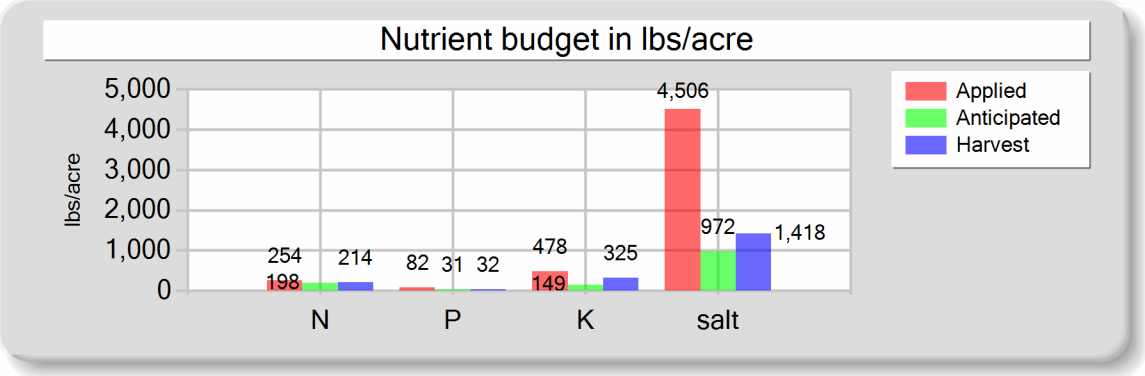
Fresh water applied	
21,288,000.00	*gallons*
783.96	*acre-inches*
30.15	*inches/acre*

Process wastewater applied	
1,152,000.00	*gallons*
42.42	*acre-inches*
1.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/08/2022: Wheat, silage, soft dough

Field name: Field 4 Crop: Wheat, silage, soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.45	49.59	146.87	2,676.74
Process wastewater	82.99	32.01	330.82	1,586.22
Fresh water	0.36	0.00	0.00	243.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.79	81.59	477.68	4,506.42
Anticipated crop nutrient removal	198.00	30.60	149.40	972.00
Actual crop nutrient removal	214.26	32.25	324.64	1,417.57
Nutrient balance	39.53	49.35	153.05	3,088.85
Applied to removed ratio	1.18	2.53	1.47	3.18

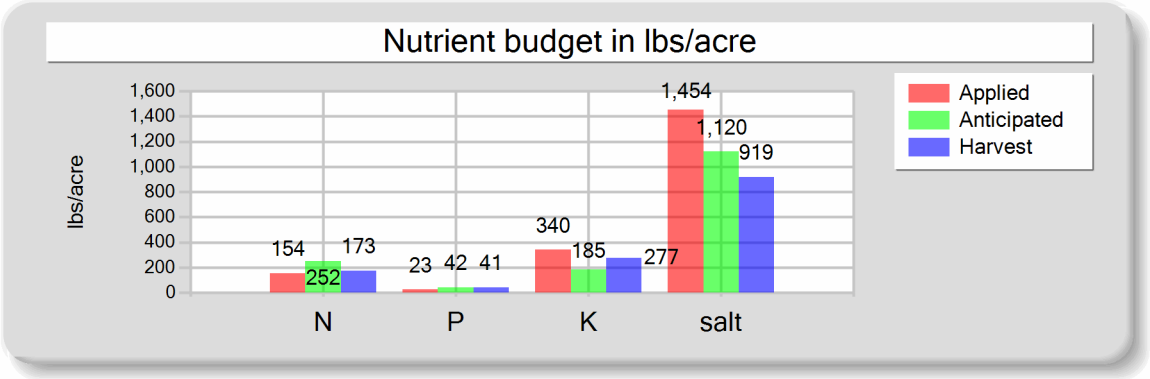
Fresh water applied	
10,320,000.00	*gallons*
380.05	*acre-inches*
7.45	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
2.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/04/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	66.59	23.24	340.15	916.10
Fresh water	0.85	0.00	0.00	538.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.43	23.24	340.15	1,454.46
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	173.34	40.59	277.05	918.55
Nutrient balance	-18.90	-17.35	63.10	535.92
Applied to removed ratio	0.89	0.57	1.23	1.58

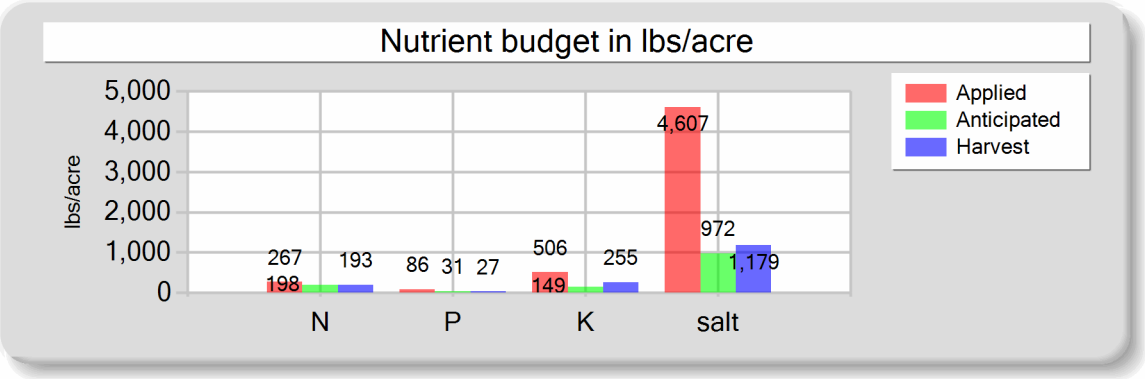
Fresh water applied	
41,400,000.00	*gallons*
1,524.62	*acre-inches*
29.89	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
1.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/08/2022: Wheat, silage, soft dough

Field name: Field 5 Crop: Wheat, silage, soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.45	49.59	146.87	2,676.74
Process wastewater	95.32	36.36	358.76	1,562.88
Fresh water	0.85	0.00	0.00	367.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.62	85.95	505.63	4,607.40
Anticipated crop nutrient removal	198.00	30.60	149.40	972.00
Actual crop nutrient removal	193.44	27.17	255.16	1,179.09
Nutrient balance	73.18	58.78	250.47	3,428.32
Applied to removed ratio	1.38	3.16	1.98	3.91

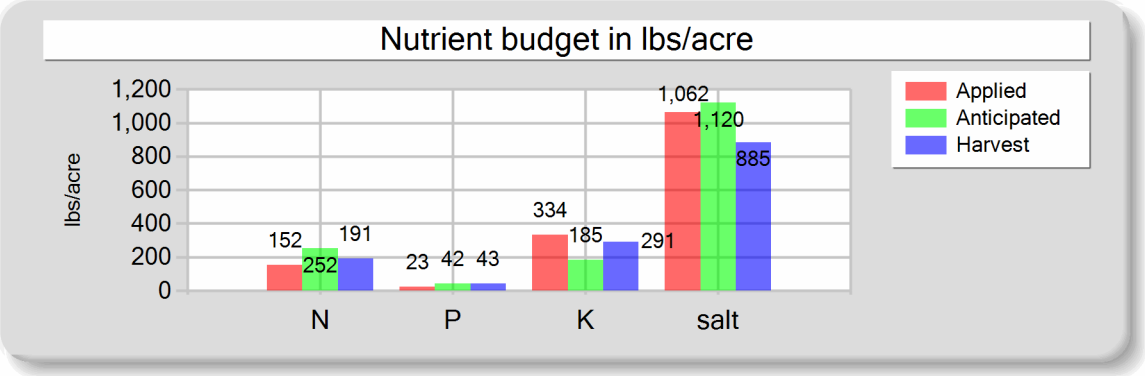
Fresh water applied	
9,000,000.00	*gallons*
331.44	*acre-inches*
6.37	*inches/acre*

Process wastewater applied	
4,120,000.00	*gallons*
151.73	*acre-inches*
2.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 06/06/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	65.31	22.79	333.60	898.49
Fresh water	0.00	0.00	0.00	163.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	152.31	22.79	333.60	1,062.03
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	191.14	42.81	290.66	884.61
Nutrient balance	-38.83	-20.01	42.95	177.42
Applied to removed ratio	0.80	0.53	1.15	1.20

Fresh water applied	
40,764,000.00	*gallons*
1,501.20	*acre-inches*
28.87	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
1.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids - 2022

Sample and source description: Corral Solids - 2022

Sample date: 09/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	6,280.00	18,600.00							33.90
DL	1,000.00	1,000.00	1,000.00							0.10

Corral Solids

Sample and source description: Corral Solids

Sample date: 06/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 27.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	4,690.00	17,400.00							0.00
DL	1,000.00	1,000.00	1,000.00							0.10

LG-Solids

Sample and source description: LG-Solids

Sample date: 06/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	5,240.00	4,910.00							0.00
DL	1,000.00	1,000.00	1,000.00							0.10

Manures

Sample and source description: Manures

Sample date: 11/07/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 26.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	10,600.00	63,000.00							0.00
DL	1,000.00	1,000.00	1,000.00							0.10

B. PROCESS WASTEWATER ANALYSES

WWS-2-4, 2022

Sample and source description: WWS-2-4, 2022

Sample date: 11/17/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	151.00	98.60	0.00	0.00	57.70	664.00								3,770.00	3,080
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

WWS-2-1

Sample and source description: WWS-2-1

Sample date: 03/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	134.00	53.20	0.00	0.00	53.30	348.00								3,060.00	1,980
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

WWS-3-1

Sample and source description: WWS-3-1

Sample date: 03/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	92.40	25.20	0.00	0.00	43.00	401.00								2,890.00	2,960
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

WWS-2-2

Sample and source description: WWS-2-2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	157.00	28.00	0.00	0.00	54.80	802.00								3,350.00	2,160
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

WWS-2-3

Sample and source description: WWS-2-3

Sample date: 08/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	95.20	50.40	0.00	0.00	47.60	432.00								2,480.00	1,740
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

WWS-2-4

Sample and source description: WWS-2-4

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	165.00	96.80	0.00	0.00	71.90	618.00								3,710.00	2,520
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

C. FRESH WATER ANALYSES

Dom-1

Dom-1

Sample description: Dom-1

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.67	0.00	0.67								443.00	284
DL	0.14	0.50	0.14								1.00	10

Dom-2

Dom-2

Sample description: Dom-2

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.25	0.00	0.25								467.00	299
DL	0.14	0.50	0.14								1.00	10

Dom-3

Dom-3

Sample description: Dom-3

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								327.00	209
DL	0.14	0.50	0.14								1.00	10

IW-1

IW-1

Sample description: IW-1

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.50	0.00	0.50								273.00	175
DL	0.14	0.50	0.14								1.00	10

IW-2

IW-2

Sample description: IW-2

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.69	0.00	0.69								323.00	206
DL	0.14	0.50	0.14								1.00	10

IW-3

IW-3

Sample description: IW-3

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.49	0.00	0.49								471.00	302
DL	0.14	0.50	0.14								1.00	10

MID Canal

MID Canal

M.I.D.

Sample description: M.I.D.

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.10	25
DL	0.14	0.50	0.14								0.10	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/07/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,200.00	3,040.00	2,970.00		14.20
DL	1,000.00	1,000.00	1,000.00		0.10

Field 1 - 05/30/2023: Corn, silage

Field 1 - 05/30/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	1,800.00	16,800.00		9.50
DL	1,000.00	1,000.00	1,000.00		0.10

Field 2 - 11/07/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	2,710.00	26,400.00		14.50
DL	1,000.00	1,000.00	1,000.00		0.10

Field 2 - 06/01/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	1,670.00	15,300.00		9.40
DL	1,000.00	1,000.00	1,000.00		0.10

Field 3 - 11/08/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	2,900.00	28,100.00		13.10
DL	1,000.00	1,000.00	1,000.00		0.10

Field 3 - 06/01/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	1,830.00	15,100.00		10.40
DL	1,000.00	1,000.00	1,000.00		0.10

Field 4 - 11/08/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	2,980.00	30,000.00		13.10
DL	1,000.00	1,000.00	1,000.00		0.10

Field 4 - 06/04/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 09/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	2,740.00	18,700.00		6.20
DL	1,000.00	1,000.00	1,000.00		0.10

Field 5 - 11/08/2022: Wheat, silage, soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,000.00	2,950.00	27,700.00		12.80
DL	1,000.00	1,000.00	1,000.00		0.10

Field 5 - 06/06/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 09/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,710.00	18,400.00		5.60
DL	1,000.00	1,000.00	1,000.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

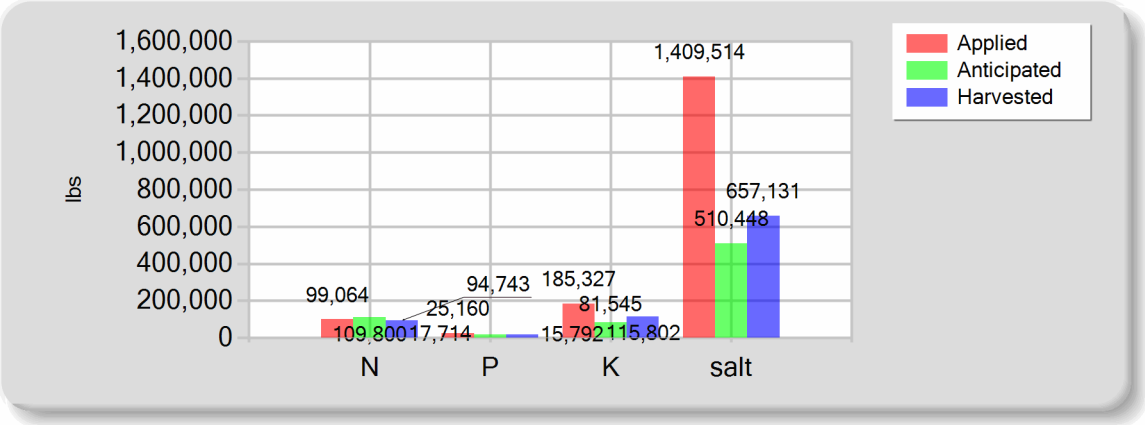
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

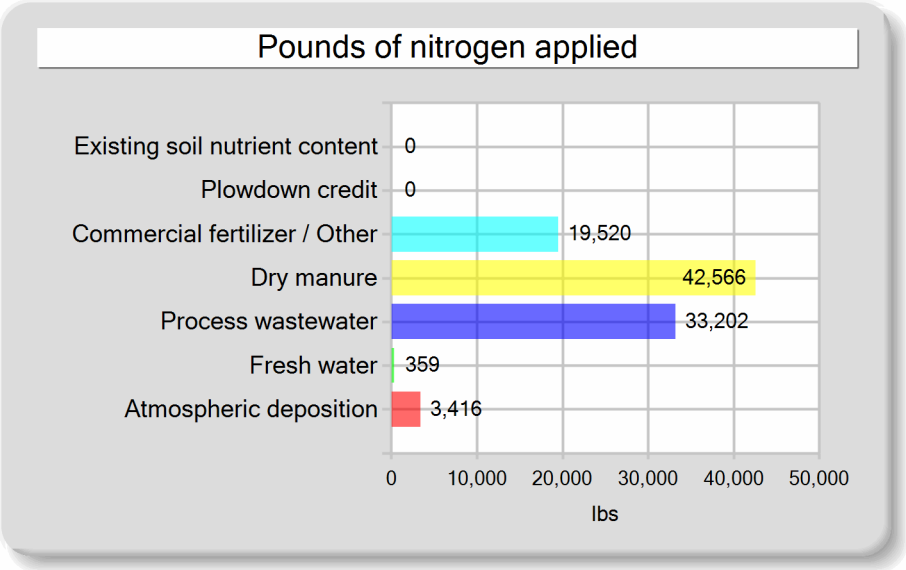
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

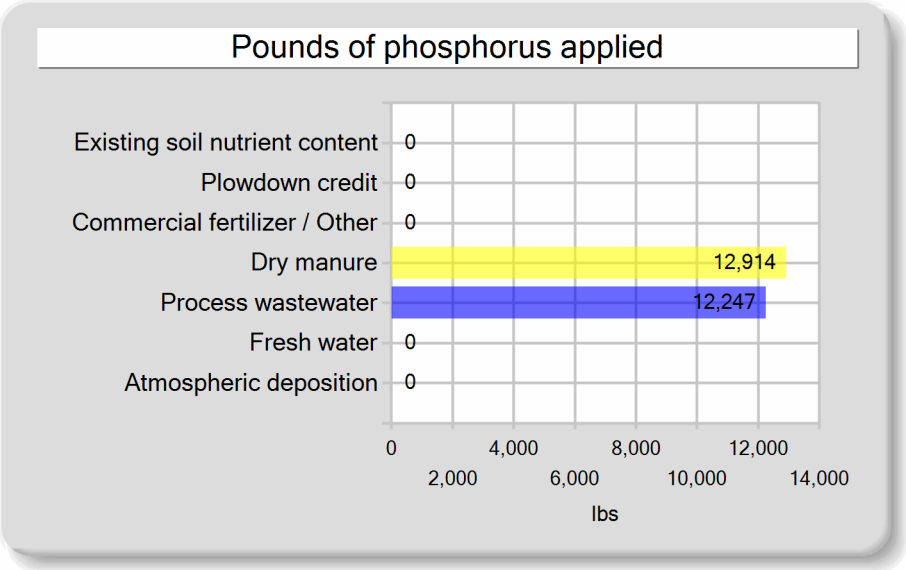
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	19,520.00	0.00	0.00	0.00
Dry manure	42,566.32	12,913.84	38,248.00	697,100.62
Process wastewater	33,202.28	12,246.53	147,079.22	538,532.89
Fresh water	359.25	0.00	0.00	173,880.74
Atmospheric deposition	3,416.00	0.00	0.00	0.00
Total nutrients applied	99,063.86	25,160.37	185,327.22	1,409,514.25
Anticipated crop nutrient removal	109,800.00	17,714.40	81,544.80	510,448.00
Actual crop nutrient removal	94,743.39	15,792.33	115,801.78	657,130.78
Nutrient balance	4,320.47	9,368.03	69,525.44	752,383.47
Applied to removed ratio	1.05	1.59	1.60	2.14

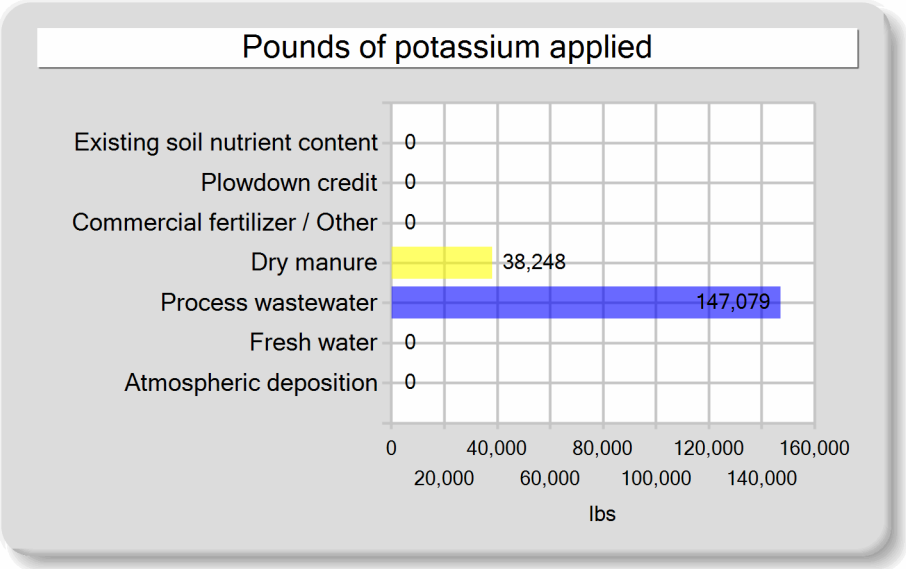
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

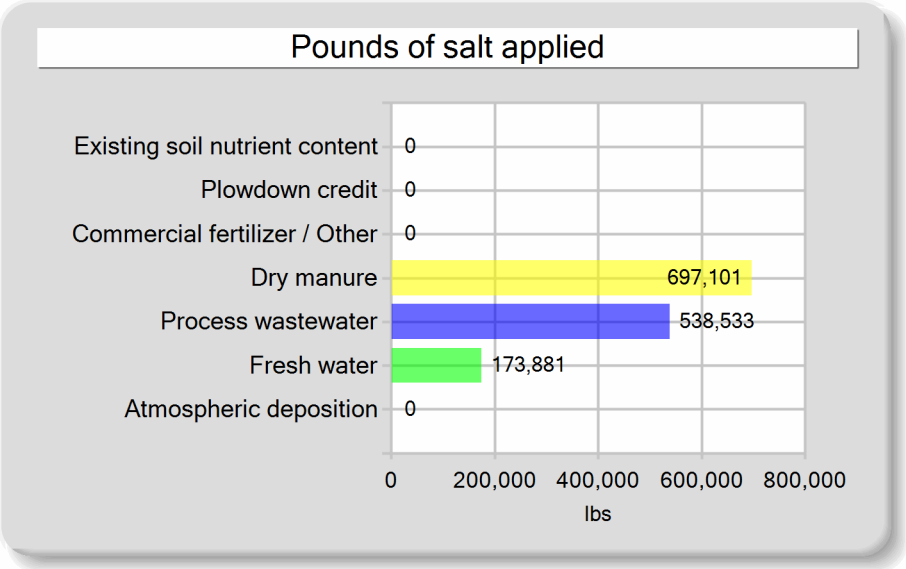


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Eugenio Azevedo Dairy #4:

No milk cows were present during the calendar year. Dry Cows and heifers were moved in and out at different times, varing the average counts. No barn water was produced. Flush was utilized on a limited basis in free stall barns with Dry Cows and some Heifers.

Lagoon water applications were hold over storage and current rainfalls. Fresh irrigation waters were added to Lagoon to dilute applications.

Additional Solid Manures was held in storage as bedding and in corrals into the 2024 Calendar year.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Tony J Machado (Sep 18, 2024 16:06 PDT)	Eugenio Azevedo (Sep 20, 2024 09:21 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Tony Machado	Eugenio Azevedo
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Sep 18, 2024	Sep 20, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Eugenio Azevedo Dairy
6178 Spring Valley Dr.
Atwater, CA 95301

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER: E & D Azevedo Dairy #4
RECEIVED DATE: 8/17/23 2:40 PM
SUBMITTED BY: Dan
ANALYZED DATE: 8/18/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW-1	T3229031B	Electrical Conductivity (EC)	EPA 120.1 and	0.273	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.503	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	175	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	91.0 (T3229026A)	94.6 (T3229026A)	3.87
Electrical Conductivity (EC)	100			1.27
Nitrate Nitrogen (NO3-N)	103	99.7 (T3229026A)	97.3 (T3229026A)	2.46

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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PURCHASE ORDER: E & D Azevedo Dairy #4
RECEIVED DATE: 8/17/23 2:40 PM
SUBMITTED BY: Dan
ANALYZED DATE: 8/18/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW-2	T3229031C	Electrical Conductivity (EC)	EPA 120.1 and	0.323	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.689	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	206	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.4	96.4 (T3229031C)	99.4 (T3229031C)	3.06
Electrical Conductivity (EC)	100			1.27
Nitrate Nitrogen (NO3-N)	105	94.6 (T3230020A)	93.6 (T3230020A)	0.978

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Atwater, CA 95301

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER: E & D Azevedo Dairy #4
RECEIVED DATE: 8/17/23 2:40 PM
SUBMITTED BY: Dan
ANALYZED DATE: 8/18/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
M.I.D.	T3229031D	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	25.2	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.4	96.4 (T3229031C)	99.4 (T3229031C)	3.06
Electrical Conductivity (EC)	100			1.27
Nitrate Nitrogen (NO3-N)	105	94.6 (T3230020A)	93.6 (T3230020A)	0.978

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: E&D AZEVEDO DAIRY #4
Mailing address: 6178 Spring Valley Dr. Atwater CA 95301
(not facility address)
Phone#: 209-404-3973

Authorized Copy Released To:
hlcdan@yahoo.com

Sampled By: Daniel Roos
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8-16-23	7:20	LG-2-3	IRRIGATION #2	21.4			X			
2	↓	7:30	IW-1		18.6		X				
3		7:40	IW-2		19.0		X				
4	8-16-23	7:55	M.I.D		19.4		X				
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st		HeartLand Consulting		8-17-23 1440		
2nd		DENELE ANALYTICAL, INC.	8/17/23 1440			T322 9031

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

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Compliance Analysis Report

Customer
Eugenio Azevedo Dairy
6178 Spring Valley Dr.
Atwater, CA 95301

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER: E & D Azevedo Dairy #4
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dom-2	T3334012A	Electrical Conductivity (EC)	EPA 120.1 and	0.467	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.253	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	299	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	96.0			0.032
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER: E & D Azevedo Dairy #4
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dom-1	T3334012B	Electrical Conductivity (EC)	EPA 120.1 and	0.443	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.673	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	284	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	96.0			0.032
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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Merced, CA 95344

PURCHASE ORDER: E & D Azevedo Dairy #4
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dom-3	T3334012C	Electrical Conductivity (EC)	EPA 120.1 and	0.327	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	209	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	96.0			0.032
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Eugenio Azevedo Dairy
6178 Spring Valley Dr.
Atwater, CA 95301

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER: E & D Azevedo Dairy #4
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW-3	T3334012D	Electrical Conductivity (EC)	EPA 120.1 and	0.471	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.488	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	302	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	96.0			0.032
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Eugenio E&D AZEVEDO DAIRY #4
Mailing address: 6178 Spring Valley Dr. Atwater CA 95301
(not facility address)
Phone#: 209-404-3973

Authorized Copy Released To: hlcdan@yahoo.com

Sampled By: Daniel Roos
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/30/23	810	DOM-2		15.6°C		X				
2	↓	815	DOM-1		15.2°		X				
3	↓	825	DOM-3		15.6°		X				
4	11-30-23	830	IW-3		15.6°		X				
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	[signature]	HeartLand Consulting		11-30-23 1405		
2nd	[signature]	DENELE ANALYTICAL, INC.	11-30-23 2:05			T3334012

Required to be submitted to any agency? (YES / NO)

2023 AR E&D AZEVEDO #4

Final Audit Report 2024-09-20

Created:	2024-09-18
By:	Daniel Roos (hlcdan@yahoo.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAncp-L9v6DRqnzNsDBxelttuxphHnUVi0

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